Quadratic stabilization of the canonical Foulkes–Howe map

Rewritten for readability from the OpenAI manuscript of the same title

Abstract

For every finite-dimensional complex vector space V, the canonical Foulkes–Howe map $\mathrm{Sym}^b(\mathrm{Sym}^a V) \to \mathrm{Sym}^a(\mathrm{Sym}^b V)$ is surjective whenever $a \geq 2$ and $b \geq a(a-1)$; the bound does not depend on $\dim V$. This is a step-by-step rewrite of the source's proof, with statements, constants and strategy unchanged.

1 The result

Setting. Let V be a finite-dimensional complex vector space and $\mathrm{Sym}\, V = \bigoplus_{j\geq 0} \mathrm{Sym}^j V$ its symmetric algebra. Products of vectors are always taken in $\mathrm{Sym}\, V$. Fix an integer $a \geq 1$. The algebra $\mathcal{P} = (\mathrm{Sym}\, V)^{\otimes a}$ is commutative under slotwise multiplication. (The source calls it P_a; it is renamed because P will denote a product of vectors.) The symmetric group $\mathfrak{S}_a$ permutes the tensor slots, acting by algebra automorphisms. Put

$$R_j = \big((\mathrm{Sym}^j V)^{\otimes a}\big)^{\mathfrak{S}_a}, \qquad R = \bigoplus_{j\geq 0} R_j.$$

Invariants of algebra automorphisms are closed under multiplication, and $R_i R_j \subseteq R_{i+j}$, so R is a graded commutative algebra.

Averaging. For any complex vector space W there is a linear isomorphism $\mathrm{Sym}^a W \to (W^{\otimes a})^{\mathfrak{S}_a}$,

$$w_1 \cdots w_a \longmapsto \mathrm{Av}(w_1 \otimes \cdots \otimes w_a) = \frac{1}{a!} \sum_{\pi \in \mathfrak{S}_a} w_{\pi(1)} \otimes \cdots \otimes w_{\pi(a)}.$$

It is well defined because the right side is symmetric in the w_i. Its inverse is the quotient projection $W^{\otimes a} \to \mathrm{Sym}^a W$ restricted to invariant tensors (standard, since $a! \neq 0$). With $W = V$ and $W = \mathrm{Sym}^b V$ this identifies $R_1 = \mathrm{Sym}^a V$ and $R_b = \mathrm{Sym}^a(\mathrm{Sym}^b V)$.

The map. For $b \geq 0$, multiplication in R gives a linear map $\mathrm{Sym}^b R_1 \to R_b$, $z_1 \cdots z_b \mapsto z_1 \cdots z_b$. Under the identifications above this is the *canonical Foulkes–Howe map*

$$\mu_{a,b,V} : \mathrm{Sym}^b(\mathrm{Sym}^a V) \longrightarrow \mathrm{Sym}^a(\mathrm{Sym}^b V). \tag{1}$$

Its image is A_b, the span of all products of b elements of R_1. This is the degree-b part of the subalgebra $A \subseteq R$ generated by R_1.

Theorem 1. *For every integer $a \geq 2$, every integer $b \geq a(a-1)$, and every finite-dimensional complex vector space V, the map $\mu_{a,b,V}$ in (1) is surjective. Equivalently, $A_b = R_b$.*

For $a = 1$ the map is the identity in every degree.

Proof at a glance. *Key idea: dualize.* Section 2 shows that $\mu_{a,b,V}$ is surjective if and only if every symmetric a-linear form T on $\operatorname{Sym}^b V$ with $T(P,\dots,P)=0$ for all products $P=u_1\cdots u_b$ of b vectors is zero. Proposition 3 proves that such a T is zero. The proof is by induction on a, with $r=a-1$ and $m=b-r$, in four steps.

1. *Binary form.* Since x^r+ct^r factors into linear factors, $(x^r+ct^r)Q$ is a product of b vectors for every product Q of m vectors. The coefficient of c gives $T(x^rQ,\dots,x^rQ,t^rQ)=0$.
2. *Freeing the last slot.* Starting from $T(x^rQ,\dots,x^rQ,t^b)$, m derivative operators each trade one factor t for a factor y_i of Q and arrive at the expression of Step 1. Lemma 2 says each operator is injective, so $T(x^rQ,\dots,x^rQ,t^b)=0$.
3. *Induction.* For fixed x,t, the form $(Q_1,\dots,Q_r)\mapsto T(x^rQ_1,\dots,x^rQ_r,t^b)$ is symmetric and r-linear on $\operatorname{Sym}^m V$, and by Step 2 its diagonal vanishes on products. The induction hypothesis shows it is zero.
4. *Removing x^r.* Starting from $T(v_1^b,\dots,v_r^b,t^b)$, r^2 operators each trade one factor v_i for a factor of one shared variable x and arrive at an expression that is zero by Step 3. Lemma 2 again gives injectivity, so T vanishes on pure powers, which span.

Dependencies. Lemma 2 feeds Steps 2 and 4. Each step uses the previous one. The spanning fact (2.1) feeds the criterion, the base case and the end of Step 4. *Where the bound enters.* The hypothesis $b\ge a(a-1)=r(r+1)$ is $m\ge r^2$. Step 3 uses it as $m\ge r(r-1)$. Step 4 uses it as $m+1>r^2-1$: the r^2 operators there all raise the degree of the same variable x, which is why the bound is quadratic.

Context. This paragraph reports the source's account of the literature. It is not used in the proof and has not been re-derived here. In positive dimension, Raicu–Sam–Weyman [11, Section 2] formulate the map as $\operatorname{Sym}^b(D^aV)\to D^a(\operatorname{Sym}^b V)$ with divided powers realized as invariant tensors, and averaging transports it to (1). The ring A is the coordinate ring of the affine Chow variety of products of a linear forms in $\operatorname{Sym}^a(V^*)$, with normalization R [11, Section 2, Theorem 2.3]; see also [8, Section 7.6, Propositions 7.11–7.12]. Brion proved surjectivity for all large b [2] and gave an effective bound involving both a and $\dim V$ [3, Theorem 3.3]. [11, Theorem 3.9] bounds the regularity of related modules, including the normalization. It does not compute the top nonzero degree of R/A, and that degree determines the least b_0 with $\mu_{a,b,V}$ surjective for all $b\ge b_0$. The source states that Theorem 1 answers positively the request for a polynomial bound in [8, Problem 7.19]: in that notation $d=b$, $n=a$, $w=\dim V$, and the bound is $d\ge n(n-1)$, including when $w=n$. Sharpness is not claimed. The Foulkes conjecture [5] asks, for $a\le b$, only for *some* $\mathrm{GL}(V)$-equivariant injection $\operatorname{Sym}^a(\operatorname{Sym}^b V)\hookrightarrow\operatorname{Sym}^b(\operatorname{Sym}^a V)$ [11, Conjecture 2.10]. Corollary 4 supplies one in our range. The existence of an embedding does not specify the canonical map: at $(a,b)=(5,5)$ and $(6,6)$, outside our range, the canonical map has nonzero kernel in suitable dimensions [4, arXiv version, Theorem 6(b),(c) and Remark 7]. Successive weight shifts like those of Steps 2 and 4 occur in Ikenmeyer's proof [7, Section 5] of McKay's propagation theorem [10]. That theorem concerns the canonical maps in the opposite direction with $\dim V\ge b$: for fixed a and $B\ge a$, injectivity at (a,B) propagates to every (a,b) with $b\ge B$ [7, Theorem 1.3 and Section 3].

2 The diagonal criterion

Fix integers $a,b\ge 1$ and put $E=\operatorname{Sym}^b V$. A *product of b vectors* is an element $u_1\cdots u_b\in E$ with $u_i\in V$. Repeats and zeros are allowed.

(2.1) Pure powers span. *For $k \geq 1$, the powers u^k with $u \in V$ span* $\operatorname{Sym}^k V$. This is used in (2.2), in the base case of Proposition 3 and at the end of its Step 4. It follows from the polarization identity

$$k!\, u_1 \cdots u_k = \sum_{J \subseteq \{1,\dots,k\}} (-1)^{k-|J|} \Big(\sum_{j \in J} u_j \Big)^k .$$

To check the identity, expand the right side with the u_i as formal commuting variables. A monomial whose set of occurring indices is S appears in the J-term exactly when $J \supseteq S$, each time with the same multinomial coefficient. Its total coefficient is that multinomial times $\sum_{J \supseteq S} (-1)^{k-|J|} = (1-1)^{k-|S|}$, which is zero unless $S = \{1, \dots, k\}$. The only degree-k monomial with full index set is $u_1 \cdots u_k$, with multinomial coefficient $k!$. Now substitute vectors. Products of k vectors span $\operatorname{Sym}^k V$ and $k! \neq 0$, so powers span.

(2.2) The image. Averaging sends u^a to $z(u) := u^{\otimes a}$, since all $a!$ summands are equal. By (2.1) with $k = a$, the elements $z(u)$ span R_1. Slotwise multiplication gives $z(u_1) \cdots z(u_b) = (u_1 \cdots u_b)^{\otimes a}$. Products of b elements from a spanning set of R_1 span A_b, so

$$A_b = \operatorname{span}_{\mathbb{C}} \{ P^{\otimes a} : P \text{ a product of } b \text{ vectors} \}.$$

(2.3) Functionals and forms. For $\lambda \in R_b^*$ define $T_\lambda(w_1, \dots, w_a) = \lambda\big(\operatorname{Av}(w_1 \otimes \cdots \otimes w_a)\big)$ for $w_i \in E$. This is a symmetric a-linear form on E. *Claim: $\lambda \mapsto T_\lambda$ is a bijection from R_b^* onto the symmetric a-linear forms on E.*

Onto. Given a symmetric T, let $\tilde{T} : E^{\otimes a} \to \mathbb{C}$ be the linear map with $\tilde{T}(w_1 \otimes \cdots \otimes w_a) = T(w_1, \dots, w_a)$, and let λ be its restriction to the subspace $R_b = (E^{\otimes a})^{\mathfrak{S}_a}$. Then $T_\lambda(w_1, \dots, w_a) = \frac{1}{a!} \sum_\pi T(w_{\pi(1)}, \dots, w_{\pi(a)}) = T(w_1, \dots, w_a)$ by symmetry.

One-to-one. Simple tensors span $E^{\otimes a}$ and Av fixes invariant tensors, so the elements $\operatorname{Av}(w_1 \otimes \cdots \otimes w_a)$ span R_b. Hence $T_\lambda = 0$ forces $\lambda = 0$.

(2.4) Conclusion. For a product P we have $\operatorname{Av}(P^{\otimes a}) = P^{\otimes a}$, so $T_\lambda(P, \dots, P) = \lambda(P^{\otimes a})$. By (2.2), λ vanishes on A_b exactly when the diagonal of T_λ vanishes on all products of b vectors. Since R_b is finite-dimensional, $A_b = R_b$ if and only if 0 is the only functional vanishing on A_b. With (2.3) this gives the following.

Diagonal criterion. *The map $\mu_{a,b,V}$ is surjective if and only if every symmetric a-linear form T on* $\operatorname{Sym}^b V$ *satisfying $T(P, \dots, P) = 0$ for all products P of b vectors is zero.*

3 Differential injectivity

Assume $n = \dim V > 0$ and fix a basis $e_1, \dots, e_n$ of V. A *vector variable* is a block $U = (U_1, \dots, U_n)$ of independent scalar variables. We write $u = \sum_k U_k e_k$ for the vector it parametrizes and, in Section 4, use the lower-case letter for both. For two different blocks U, Y define

$$D_{Y,U} = \sum_{k=1}^{n} Y_k \frac{\partial}{\partial U_k}.$$

It lowers the U-degree by one and raises the Y-degree by one. Section 4 uses two elementary facts.

(D1) A polynomial in finitely many variables over $\mathbb{C}$ that vanishes at every point is the zero polynomial. This follows by induction on the number of variables from the univariate root bound.

(D2) Let T be a multilinear form on $\mathrm{Sym}^b V$, let $j \geq 1$, and let one slot be $u^j w$, where w and all other slots do not involve U. Then $D_{Y,U}\, T(\ldots, u^j w, \ldots) = j\, T(\ldots, u^{j-1} y\, w, \ldots)$. Indeed, the Leibniz rule in the commutative algebra $\mathrm{Sym}\, V$ gives $\partial u^j / \partial U_k = j\, u^{j-1} e_k$. Multiply by Y_k, sum over k, and use linearity of T in that slot.

The next lemma is what allows a derivative to be cancelled. It is used in Steps 2 and 4 of Proposition 3. The source compares it with positive-weight injectivity in $\mathfrak{sl}_2$ weight strings [6, Section 11.1, Claim 11.4 and equation (11.5)] and with [7, Section 5, Claim 5.1]. The proof below uses neither.

Lemma 2. *Let p, q be nonnegative integers. On polynomials homogeneous of degree p in U and degree q in Y, the operator $D_{Y,U}$ is injective if $p > q$. The polynomials may depend on any finite collection of additional variables independent of U and Y.*

Proof. Write $D = D_{Y,U}$ and $D' = D_{U,Y}$. (The source calls the latter E; it is renamed to avoid a clash with $E = \mathrm{Sym}^b V$.) Let h have bidegree (p, q).

Step 1: inner product. Let X list all scalar variables, including the additional ones. Give the polynomial ring the Fischer (apolar) Hermitian inner product [1, Section 2, equation (8) and Proposition 3]: distinct monomials are orthogonal and $\|X^\alpha\|^2 = \alpha! := \prod_k \alpha_k!$. It is positive definite because $\alpha! > 0$.

Step 2: adjoints. For each coordinate X_k, $\langle \partial_{X_k} f, g\rangle = \langle f, X_k g\rangle$. It suffices to take $f = X^\alpha$ and $g = X^\beta$. Both sides are then $\alpha!$ if $\alpha = \beta + e_k$ and 0 otherwise, because $\alpha_k\, (\alpha - e_k)! = \alpha!$. Hence $D^* = \sum_k (\partial_{U_k})^* (Y_k)^* = \sum_k U_k \partial_{Y_k} = D'$, and likewise $(D')^* = D$.

Step 3: commutator. Using $[\partial_{Y_j}, Y_k] = \delta_{jk}$ and $[U_j, \partial_{U_k}] = -\delta_{jk}$, with all other pairs commuting,

$$[U_j \partial_{Y_j},\, Y_k \partial_{U_k}] = U_j [\partial_{Y_j}, Y_k] \partial_{U_k} + Y_k [U_j, \partial_{U_k}] \partial_{Y_j} = \delta_{jk} \big(U_j \partial_{U_j} - Y_j \partial_{Y_j} \big).$$

Summing over j, k gives $D'D - DD' = N_U - N_Y$, where $N_U = \sum_k U_k \partial_{U_k}$ and $N_Y = \sum_k Y_k \partial_{Y_k}$. These Euler operators multiply each monomial by its U-degree and Y-degree respectively, so $N_U h = ph$ and $N_Y h = qh$.

Step 4: norm identity. By Steps 2 and 3,

$$\|Dh\|^2 - \|D'h\|^2 = \langle D'Dh, h\rangle - \langle DD'h, h\rangle = \langle (N_U - N_Y)h, h\rangle = (p - q)\|h\|^2. \tag{2}$$

Step 5. If $Dh = 0$ and $p > q$, the left side of (2) is ≤ 0 and the right side is ≥ 0. So $(p-q)\|h\|^2 = 0$ and $h = 0$. □

How it is applied. To deduce $g = 0$ from $D_k \cdots D_1 g = 0$, peel operators off from the outside. This requires the degree inequality of the lemma for the bidegree of $D_{i-1} \cdots D_1 g$, for each i.

4 Vanishing on products

Proposition 3. *Let V be a finite-dimensional complex vector space, and let $a \geq 1$ and $b \geq 1$ be integers satisfying $b \geq a(a-1)$. If a symmetric a-linear form T on $\mathrm{Sym}^b V$ satisfies*

$$T(P, \ldots, P) = 0 \qquad \text{for every product } P \text{ of } b \text{ vectors}, \tag{3}$$

then $T = 0$.

Proof. Plan. We induct on a, proving the statement for all admissible b and all V at once. Steps 1–4 show that T vanishes on all tuples of pure powers. Then (2.1) finishes the proof.

If $V = 0$ then $\operatorname{Sym}^b V = 0$ because $b \geq 1$, so $T = 0$. Assume $\dim V > 0$ and use the conventions of Section 3.

Base case $a = 1$. T is a linear form vanishing on products of b vectors, which span $\operatorname{Sym}^b V$.

Inductive step. Let $a \geq 2$ and assume the statement for $a - 1$. Put

$$r = a - 1 \geq 1, \qquad m = b - r.$$

The hypothesis $b \geq a(a-1) = r(r+1)$ gives $m \geq r^2 \geq 1$.

Step 1: a split binary form. *Claim: for all $x, t \in V$ and every product Q of m vectors,*

$$T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}}, t^r Q) = 0. \tag{4}$$

(a) *For every $c \in \mathbb{C}$, $(x^r + ct^r)Q$ is a product of b vectors.* Choose $\alpha \in \mathbb{C}$ with $\alpha^r = -c$ and let $\zeta = e^{2\pi i/r}$. Homogenizing $z^r - 1 = \prod_j (z - \zeta^j)$ gives, in $\mathbb{C}[X, Y]$,

$$X^r + cY^r = \prod_{j=0}^{r-1} (X - \alpha\zeta^j Y).$$

For $c = 0$ this reads $X^r = X^r$. Since $\operatorname{Sym} V$ is commutative, there is an algebra homomorphism $\mathbb{C}[X, Y] \to \operatorname{Sym} V$ with $X \mapsto x$, $Y \mapsto t$. This needs no independence of x and t. Applying it, $x^r + ct^r = \prod_j (x - \alpha\zeta^j t)$ is a product of r vectors, and multiplying by Q gives a product of $r + m = b$ vectors.

(b) *Coefficient of c.* By (a) and (3), $T(P_c, \dots, P_c) = 0$ for all c, where $P_c = x^r Q + c\, t^r Q$. By multilinearity and symmetry,

$$T(P_c, \dots, P_c) = \sum_{k=0}^{a} \binom{a}{k} c^k\, T(\underbrace{x^r Q, \dots}_{a-k}, \underbrace{t^r Q, \dots}_{k}).$$

A polynomial in c vanishing for all c has zero coefficients. The coefficient of c is $a\, T(x^r Q, \dots, x^r Q, t^r Q)$ with $a - 1 = r$ copies of $x^r Q$, and $a \neq 0$. This proves (4).

Step 2: freeing the last slot. *Claim: for all $x, t \in V$ and every product Q of m vectors,*

$$T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}}, t^b) = 0. \tag{5}$$

Fix $x \in V$. Let $t, y_1, \dots, y_m$ be independent vector variables, put $Q = y_1 \cdots y_m$, and for $0 \leq k \leq m$ define the polynomial

$$G_k = T(x^r Q, \dots, x^r Q,\ t^{b-k} y_1 \cdots y_k).$$

The source's H_x is G_0, the left side of (5).

(a) $G_m = 0$ *as a polynomial.* Since $b - m = r$, the last slot of G_m is $t^r Q$. So G_m vanishes at every point by (4), and (D1) applies.

(b) $D_{y_k,t} G_{k-1} = (b - k + 1)\, G_k$ *for* $1 \leq k \leq m$. Only the last slot involves t. Apply (D2) with $j = b - k + 1$ and $w = y_1 \cdots y_{k-1}$. The scalar $b - k + 1$ is positive. Composing gives the source's formula $D_{y_k,t} \cdots D_{y_1,t} G_0 = \frac{b!}{(b-k)!} G_k$.

(c) *Bidegree.* G_{k-1} has degree $p = b-k+1$ in t, which occurs only in the last slot. It has degree $q = r$ in y_k: each of the first r slots contains Q, which is linear in y_k, and the last slot does not yet contain y_k. Thus $p - q = m - k + 1 \geq 1$.

(d) *Descent.* If $G_k = 0$, then $D_{y_k,t}G_{k-1} = 0$ by (b), so $G_{k-1} = 0$ by Lemma 2 and (c), with the other variables as additional variables. Starting from (a) and taking $k = m, m-1, \ldots, 1$ yields $G_0 = 0$. Evaluating at arbitrary vectors gives (5).

Step 3: induction on the remaining slots. *Claim: for all $x, t \in V$,*

$$T(x^r Q_1, \ldots, x^r Q_r, t^b) = 0 \qquad (Q_1, \ldots, Q_r \in \operatorname{Sym}^m V). \tag{6}$$

Fix x, t and let $S_{x,t}(Q_1, \ldots, Q_r)$ denote the left side. It is r-linear because $Q \mapsto x^r Q$ is a linear map $\operatorname{Sym}^m V \to \operatorname{Sym}^b V$, and symmetric because T is. By (5), its diagonal vanishes at every product of m vectors. The induction hypothesis, for r slots and degree m, requires $m \geq 1$ and $m \geq r(r-1)$. Both follow from $m \geq r^2 \geq 1$. Hence $S_{x,t} = 0$.

Step 4: removing the common power. *Claim: $T(v_1^b, \ldots, v_r^b, t^b) = 0$ for all $v_1, \ldots, v_r, t \in V$.* Fix $t \in V$. Let $v_1, \ldots, v_r, x$ be independent vector variables. For $s = (s_1, \ldots, s_r)$ with $0 \leq s_i \leq r$ put $|s| = \sum_i s_i$ and

$$F_s = T(x^{s_1} v_1^{b-s_1}, \ldots, x^{s_r} v_r^{b-s_r},\ t^b).$$

The source's f is $F_{(0,\ldots,0)}$.

(a) $F_{(r,\ldots,r)} = 0$ *as a polynomial.* It equals $T(x^r v_1^m, \ldots, x^r v_r^m, t^b)$. This vanishes at every point by (6) with $Q_i = v_i^m$, and (D1) applies.

(b) *If $s_i < r$ then $D_{x,v_i} F_s = (b - s_i)\, F_{s+e_i}$,* where e_i is the i-th unit vector. Only slot i involves v_i. Apply (D2) with $j = b - s_i$ and $w = x^{s_i}$. The scalar $b - s_i \geq m + 1$ is positive. Applying D^r_{x,v_i} for $i = 1, \ldots, r$ in turn gives the source's formula $D^r_{x,v_r} \cdots D^r_{x,v_1} F_{(0,\ldots,0)} = \left(\frac{b!}{m!}\right)^r F_{(r,\ldots,r)}$.

(c) *Bidegree.* F_s has degree $p = b - s_i$ in v_i and degree $q = |s|$ in x. All operators raise the degree of the same variable x. If $s_i \leq r - 1$, then $|s| \leq r^2 - 1$ and

$$p = b - s_i \ \geq\ b - r + 1 = m + 1 \ \geq\ r^2 + 1 \ >\ r^2 - 1 \ \geq\ |s| = q.$$

(d) *Descent.* If $s_i < r$ and $F_{s+e_i} = 0$, then $D_{x,v_i} F_s = 0$ by (b), so $F_s = 0$ by Lemma 2 and (c). Start from $(r, \ldots, r)$ by (a) and lower one coordinate at a time, in r^2 steps that undo the source's composition from the outside. This gives $F_{(0,\ldots,0)} = 0$. Evaluating at arbitrary vectors, for each t, proves the claim.

Conclusion. By (2.1) with $k = b$, pure powers span $\operatorname{Sym}^b V$. By Step 4 and multilinearity, $T = 0$. This completes the induction. □

Proof of Theorem 1. For $a \geq 2$ and $b \geq a(a-1)$ we have $b \geq 1$. Proposition 3 then gives the vanishing statement in the diagonal criterion of Section 2, and the criterion gives surjectivity. □

5 Representation comparison and scope

Corollary 4. *For every finite-dimensional complex vector space V, every integer $a \geq 2$, and every integer $b \geq a(a-1)$, there exists a $\mathrm{GL}(V)$-equivariant embedding*

$$\operatorname{Sym}^a(\operatorname{Sym}^b V) \hookrightarrow \operatorname{Sym}^b(\operatorname{Sym}^a V).$$

In particular, this gives the sixth symmetric-power comparison for $b \geq 30$.

The source does not define the phrase in the last sentence. We read it as the case $a = 6$, where $a(a-1) = 30$: an embedding $\mathrm{Sym}^6(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^6 V)$ for $b \geq 30$.

Proof. $\mathrm{GL}(V)$ acts on $\mathrm{Sym}\, V$ by algebra automorphisms, hence on $\mathcal{P}$, commuting with $\mathfrak{S}_a$. So averaging and multiplication are equivariant, and the surjection $\mu_{a,b,V}$ of Theorem 1 is equivariant. Finite-dimensional polynomial representations over $\mathbb{C}$ are completely reducible [9, Chapter I, Appendix A, Sections 7–8]. Choose an invariant complement M to $\ker \mu_{a,b,V}$. The restriction $\mu_{a,b,V}|_M$ is an equivariant isomorphism onto $\mathrm{Sym}^a(\mathrm{Sym}^b V)$. Its inverse followed by the inclusion $M \subseteq \mathrm{Sym}^b(\mathrm{Sym}^a V)$ is the required embedding. This shows existence only. It does not identify the embedding with a prescribed canonical map. □

Remark 5. Nothing in the proof depends on $\dim V$. Lemma 2 holds for any number of coordinates and additional variables, and every degree comparison involves only a and b. So no reduction to a particular dimension is needed. The proof uses positivity of the Fischer inner product, splitting of binary forms over $\mathbb{C}$, and the nonvanishing in characteristic zero of the scalars $k!$, a, $b!/(b-k)!$ and $(b!/m!)^r$.

Verification checklist

1. *Dualization.* The image is spanned by the $P^{\otimes a}$ (2.2), and $\lambda \mapsto T_\lambda$ is a bijection with $T_\lambda(P, \dots, P) = \lambda(P^{\otimes a})$ (2.3, 2.4).
2. *Lemma 2.* Check the adjointness $D^* = D'$ (Step 2), the commutator $D'D - DD' = N_U - N_Y$ (Step 3), and that h is homogeneous separately in U and in Y wherever the lemma is invoked.
3. *Step 1(a).* $(x^r + ct^r)Q$ really is a product of b vectors, for dependent x, t and for $c = 0$ as well. In Step 1(b) the coefficient of c has exactly r copies of $x^r Q$.
4. *Step 2(c).* Before the k-th operator the y_k-degree is still r, not $r+1$, and the t-degree is $b-k+1$, so $p - q = m - k + 1 \geq 1$.
5. *Step 3.* The induction hypothesis is applied to $(a-1, m, V)$. Its requirements $m \geq 1$ and $m \geq r(r-1)$ hold.
6. *Step 4(c).* With one shared variable x, the chain $b - s_i \geq m + 1 \geq r^2 + 1 > r^2 - 1 \geq |s|$ holds before each of the r^2 operators. Together with Step 3, this is where $b \geq a(a-1)$ is used.
7. *Passing between values and polynomials.* (D1) is used in Steps 2(a) and 4(a), and the descents then return to values at arbitrary vectors.

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